AKHIL SAGARAN KASTURI

EDUCATION

Georgia Institute of Technology, *Atlanta, GA* **Aug 2026 – Present**
- Master of Science in Computer Science specializing in Artificial Intelligence

Purdue University*, West Lafayette, IN* **Aug 2022 –Dec 2025**
- Bachelor of Science in Computer Science and Applied Statistics (Minor: Mathematics) specializing in Machine intelligence
- CGPA: 3.74/4.0
- Recipient of Dean's List and Semester Honors (Fall '22, Spring '23, Fall '23, Spring '24, Fall '24, Spring '25, Fall '25)

WORK EXPERIENCE

AI Engineer & Researcher, *PwC, Boston, MA* **Jan 2026 – Present**
- Researched and developed new architectures, tools and optimization strategies for agentic AI frameworks and multi-agent systems, translating emerging AI research into scalable enhancements for enterprise AI pipelines.
- Designed and optimized retrieval, knowledge-resolution, and contextual reasoning workflows to improve agent grounding, information synthesis, and decision quality across knowledge environments, helping achieve agentic wisdom.
- Conducted applied research on agent harnesses and information-retrieval systems and published white papers to research conferences and journals of the same.

Artificial Intelligence & Data Analytics Intern, *Crowe LLP, Chicago, IL* **Jun 2025 – Aug 2025**
- Designed and implemented a multi-agent Data Quality Analysis system leveraging LLMs to generate quality checks from metadata, convert them to YAML, Jinja templates, and SQL, and dynamically update table/column descriptions based on results; built supporting UI, testing harness, and debugging tools.
- Led R&D on LLM memory and context management modules, benchmarking standalone memory layers and developing scaling laws; findings integrated into Crowe's internal knowledge base through two technical whitepapers.
- Developed a custom agentic RAG pipeline using Semantic Kernel with modular retrieval and generation skills, featuring iterative context engineering for improved relevance and control.

Software Development Intern*, CGI, Westerville, OH* **May 2024 – Aug 2024**
- Developed and implemented new features for an internal member forecasting and progress monitoring application for NiSource using .NET technologies (C#, React) and MySQL.
- Collaborated with .NET developers to provide patch fixes and enhance application performance.
- Researched and developed AI models, including a chatbot and a multi-variable time-series model, using Python, C#, and Azure's AI and ML Studios (AI Bot Service, AI Language, AutoML) to improve member forecasting and time planning.

Undergraduate Researcher, *Purdue University, West Lafayette, IN* **May 2025 – Dec 2025**
- Researching on the possibilities of utilizing image processing techniques to transcribe speech with Dr. Abulhair Saparov. Involves developing a multi-modal pipeline that is capable of blind segmenting audio spectrograms, labelling individual segments to phonemes using visual models, and finally transcribing the audio.
- Supported clinical AI research focused on automating triage recommendations for HIV patients using LLMs to understand their reasoning abilities and scaling laws with a long-term goal of scalable, safe decision support in under-resourced care settings.

Resident Assistant*, Purdue University Residences, West Lafayette, IN* **Aug 2024 – Dec 2025**
- Employed an annual shared budget of $96,000 to coordinate unique events promoting diversity, mental health, financial wellbeing, and career growth.
- Maintained personal relationships and conducted quarterly checkups to foster a safe, healthy, and engaged community of 50+ students from 20+ states and 5 countries.
- Provided on-call response to concerns a resident, co-worker or supervisor had regarding the residence hall environment.

Undergraduate Teaching Assistant*, Purdue University, West Lafayette, IN* **Jan 2024 – Dec 2025**
- Supported 800+ students by conducting weekly labs and office hours, addressing course concepts and project queries, and providing guidance on coding standards.
- Enhanced student engagement on EdStem communication portal by promptly responding to inquiries and maintain clear communication on project deadlines.
- Assisted in course development by facilitating quiz logistics, grading, and contributing to the creation and testing of all student assignments.

PUBLICATIONS

- ***Is grep all you need? (2026):*** Co-authored an empirical study of agentic retrieval architectures, benchmarking grep and vector search across multiple LLM agent harnesses and retrieval interfaces; found that harness and tool-calling design significantly influence retrieval accuracy, with grep outperforming vector search across evaluation configurations.

PERSONAL PROJECTS

- ***Codeminder (2025):*** Developed an MCP server that pre-indexes codebases with AST-aware chunking and local embeddings, exposing semantic search tools for agents/IDEs. Added eval harnesses, chunking comparisons, and MCP integrations to accelerate retrieval and developer/agent workflows. Integrated and test performance with Codex CLI and Gemini CLI.
- ***ShellFolio (2025):*** Developed a Next.js-based Unix inspired personal portfolio, featuring interactive commands, flag-based navigation, reverse-i-search, tab completion via custom trie, and email redirection. Engineered a no-scroll UI with Resend API integration for real-time terminal-style messaging.

LANGUAGES: Python, Java, C, C++, C#, JavaScript, SQL, Shell, R
FRAMEWORKS: .NET, NodeJS, ReactJS, MySQL, MongoDB, Web Dev (HTML, CSS, jQuery), Git